MARK SCHEME · TEACHER COPY

12BR · Topic 20: Further Programming

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Total marks: 37

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Total: 37 marks

Question 1 · Oct/Nov 2025 Paper 31, original Q11 (a),(b) [8 marks]

Question	Answer	Marks

1(a)	**One** mark per mark point
	MP1 Two correct attributes (`PatientID : INTEGER` and `Doctor : STRING`)
	MP2 `SetTreatments(…)` and `SetMedications(…)` seen
	MP3 … and appropriate parameters, with string data types
	MP4 `GetPatientID()` and `GetDoctor()` seen
	MP5 `SetDateSeen(…)` and `GetDateSeen` fully correct with appropriate parameter and correct data type in setter.
	
	Appointment
	`DateSeen : DATE`
	`PatientID : INTEGER`
	`Doctor : STRING`
	`Treatments : STRING`
	`Medications : STRING`
	
	`SetDateSeen(NewDate : DATE)`
	`SetPatientID(PatientNumber : INTEGER)`
	`SetDoctor(DoctorID : STRING)`
	`SetTreatments(NewTreatments : STRING)`
	`SetMedications(NewMedications : STRING)`
	`GetDateSeen()`
	`GetPatientID()`
	`GetDoctor()`
	`GetTreatments()`
	`GetMedications()`
		5

1(b)(i)	Encapsulation	1
1(b)(ii)	**One** mark for each mark point (**Max 2**) MP1 Inheritance is where a derived class takes properties / behaviours attributes / methods MP2 … of a parent / super class / base class MP3 The attributes / methods / properties taken from the parent / super class / base class can also be extended / copied / used / changed / overwritten / overridden in the subclass.	2

Question 2 · May/June 2026 Paper 32, original Q11 (a),(b) [9 marks]

Question	Answer	Marks	Guidance

2(a)	**One** mark per mark point (**Max 5**)
	MP1 Correct `CLASS Node` and `ENDCLASS` statements
	MP2 Correct attribute declaration section before the constructor
	MP3 Correct statement to begin and end the constructor as `PUBLIC PROCEDURE NEW ... ENDPROCEDURE`
	MP4 Two parameters identified in the constructor with appropriate names and data types.
	MP5 Values passed using the constructer are assigned correctly – allow use of setters.
	
	Example algorithm
	
	`CLASS Node`
	`  PRIVATE Content : STRING`
	`  PRIVATE Pointer : INTEGER`
	
	`  PUBLIC PROCEDURE NEW(NewContent : STRING,`
	`  NewPointer : INTEGER)`
	`  Content ← NewContent`
	`  Pointer ← NewPointer`
	`  ENDPROCEDURE`
	`ENDCLASS`
		5	MP1 `Node` is given in the question, so must be used.
			MP2 must use `Content` and `Pointer` as `STRING` and `INTEGER`, respectively.
			MP4 any names can be used except `Content` and `Pointer`.
			MP5 the candidate's parameters from MP4 must be assigned to the attributes from MP2.

2(b)	**One** mark per correct answer (**Max 4**)
	
	OOP Term / **Description**
	setter — A method used to change values in private attributes in a class.
	instance(s) // object(s) — An occurrence of a class during the execution of a program.
	aggregation — One class/object contains another // Hierarchy of nested classes/objects // Process by which one class can contain other classes // Represents a "has a" relationship between classes.
		4	

inheritance	Deriving new classes/objects on existing classes/objects
	// Process in which the methods and data from one class, a superclass or base class, are copied to another class, a derived class
	// Represents a hierarchy of derived classes
	// Represents an "is a" relationship between classes.

Question 3 · Oct/Nov 2025 Paper 32, original Q12 (a),(b) [8 marks]

Question	Answer	Marks
3(a)	**One** mark per mark point MP1 Two correct attributes (`DateOfBirth : Date`) and priority with a sensible name and integer data type MP2 `SetPatientID(…)` and `SetDoctorID(…)` seen MP3 … and appropriate parameters, with string data types MP4 `GetPatientID()` and `GetDateOfBirth()` seen MP5 Priority setter and getter fully correct with name matching Priority attribute, with appropriate parameter and correct data type.	5

Patient

```
PatientID       : STRING
Name            : STRING
DateOfBirth     : DATE
Priority        : INTEGER
DoctorID        : STRING
```

	SetPatientID(PatientNumber : STRING) SetName(FullName : STRING) SetDateOfBirth(DOB : DATE) **SetPriority(Urgency : INTEGER)** **SetDoctorID(DocID : STRING)** **GetPatientID()** GetName() **GetDateOfBirth()** **GetPriority()** GetDoctorID()	
3(b)(i)	Instance // instantiation	**1**
3(b)(ii)	**One** mark for each mark point MP1 Polymorphism is when methods with the same name are redefined / behave differently … MP2 … in derived / inherited classes / subclasses	**2**

Question 4 · May/June 2026 Paper 31, original Q11 (a),(b),(c) [8 marks]

Question	Answer	Marks	Guidance
4(a)	**One** mark for each correctly completed clause (**Max 3**) (25) customer(betty). (26) owns(large_tv, betty). (27) wants(bluray_player, betty).	**3**	Accept facts in any order. The device names have been given in clauses 02 and 04. Responses must match these.
4(b)	(Owner =) petrov, susan, sammy	**1**	Can be in any order

Total: 37 marks

4(c)	**One** mark per mark point (**Max 4**)
	MP1 `device(D)`
	MP2 `customer(A)`
	MP3 `owns(D, A).`
	MP4 All logic conditions / punctuation correct (accept , for `AND`)
	
	Example answers
	
	`may_buy_device(D, A)`
	`  IF device(D) AND customer(A) AND NOT owns(D, A).`
	
	`may_buy_device(D, A)`
	`  IF device(D), customer(A), NOT owns(D, A).`
		4	Allow MP1, MP2 and MP3 in any order.
			
			Accept \+ for NOT operator as Swish Prolog.

Question 5 · Oct/Nov 2025 Paper 33, original Q5 [4 marks]

Question	Answer	Marks
5	**One** mark per correct answer (**Max 4**)	**4**

OOP Term	Purpose
Getter	A method that accesses the value of a property
Setter	A method that changes the value of a property
Object	**An instantiation / instance of a class**
Method	**A programmed function / procedure / subroutine / subprogram defined as part of a class**

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